

Power stage development for a dual active bridge DC-DC power converter

A

Thesis

Submitted in partial satisfaction of the requirements for the

Degree of

MASTER OF TECHNOLOGY

Submitted by

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November 2025

Abstract

The power stage of a dual active bridge converter transfers energy between two DC ports. It consists of two full H-bridges, a high-frequency transformer that provides galvanic isolation, and the transformer's leakage inductance, which is used intentionally as the energy transfer path. One bridge produces a square voltage on the primary, the other does the same on the secondary, and the difference across the leakage inductance shapes the inductor current that carries power between the ports.

The control begins with single phase shift modulation, where there will be a relative phase shift between the two active bridges. It is simple but encourages circulating current. Dual phase shift modulation adds a second, inner phase shift inside each bridge, creating brief zero-level intervals in the bridge voltage. Those intervals let the controller represent the inductor current, shorten the non-productive circulation, and widen the range over which soft switching is maintained.

Zero Voltage Switching arises naturally during dead time. The inductor current charges and discharges device capacitances so switches turn on with little or no voltage across them, reducing loss and stress. Reactive power appears and current still flows but does not contribute to net power transfer, raising RMS current.

Matlab Simulink studies simulate these behaviors. In open loop, single phase shift reproduces the expected current ramps and hits the desired power at nominal conditions, but power and output voltage drift with input or load changes. A closed loop restores the set-point by approaching the required phase shift. When the same simulation is driven with dual phase shift, the simulated waveforms show smaller current ripple around commutations, lower peaks, and a broader region where soft switching holds.

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Chapter 1

Introduction

1.1 Introduction

Electrified transport, DC microgrids, and storage-rich power systems increasingly rely on isolated, bidirectional DC-DC interfaces to move energy safely and efficiently between dissimilar voltage domains. Among available topologies, the Dual Active Bridge (DAB) has become a workhorse because it combines galvanic isolation with fixed-frequency operation and soft commutation, enabling compact magnetics, high power density, and predictable EMI behavior [1]. In its canonical form, two H-bridges synthesize 50% duty square waves; commanding a relative phase produces an average inductor current whose sign and magnitude set the direction and level of power flow. These principles are consistently reported across vendor design guides, tutorial notes, and the research literature [2].

Historically, the soft switched dual bridge family was motivated by high power applications, with early work showing that controlled resonant or soft transitions yield high power density and reduced device stress ideas that underpin modern DAB operation and its extensions [3]. Subsequent analyses formalized single phase shift (SPS) control (one phase angle between bridges) and revealed that, at light load, a significant fraction of the RMS current is reactive [4] (energy shuttled within a switching period but not delivered to the load). Dual phase shift (DPS) adding inner phase shifts within each bridge reduces that circulating component and broadens the zero voltage switching (ZVS) region, improving efficiency and easing thermal design.

[5]As rigs, microgrids, and aircraft electrification demand multiple isolated rails, the DAB concept generalizes to dual active bridge (DAB) and triple active bridge (TAB) converters,

enabling simultaneous routing among three or more ports via coordinated phase scheduling. Recent surveys and application studies summarize these trends and control options; they motivate the extensions explored in this work's future scope.

1.2 Background

A Non-Isolated DC–DC Converter is a small power-processing circuit that shifts one DC Voltage level to another DC Voltage level without using a transformer. These Converters show up everywhere modern electronics need tight voltage control and good efficiency inside phone, laptop chargers, battery powered sensors, EV subsystems, and renewable-energy hardware because they can step voltages up or down. The choice of a specific DC-DC converter depends on the applications power requirements, efficiency considerations, and the desired voltage transformation characteristics. There are several types of DC-DC converters, each with its own characteristics and applications. Some common types are buck converter, boost converter, and buck-boost converter. Those are shown in fig.1.1 and fig.1.2.

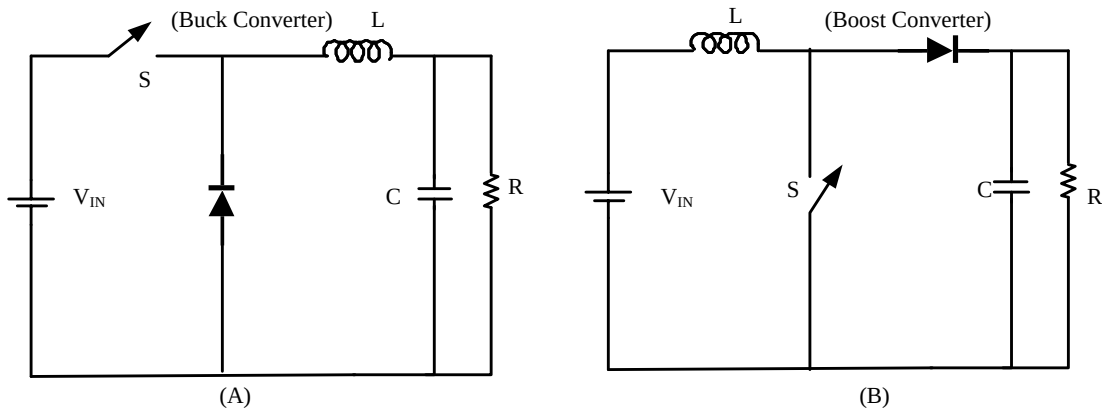


Figure 1.1: Examples of non-isolated dc-dc converters

1.2.1 Advantages

- **Voltage conversion:** DC-DC converters enable the transformation of DC voltage from one voltage level to another voltage level. This ability to step down (buck), step up (boost), or perform both functions (buck-boost) is valuable for adapting power sources to meet specific requirements in different parts of a system.

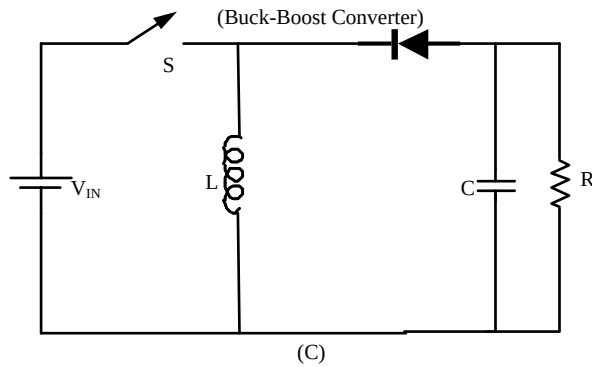


Figure 1.2: Examples of non-isolated dc-dc converters

- **voltage regulation:** DC-DC converters provide precise control over the output voltage, ensuring a stable and regulated supply.
- **Battery Management:** In battery-powered systems, DC-DC converters are contributed to effective battery management by optimizing the use of available energy and ensuring that the devices receive the required voltage levels

1.2.2 Disadvantages

- **No isolation:** General basic DC-DC converters are typically non-isolated converters, they don't provide electrical separation between input and output. If isolation is required for safety or noise reasons, a different (isolated) topology or additional components are needed.
- **Voltage Ripple:** DC-DC converters may introduce higher voltage ripple on both input and output sides in which the ripples can affect the performance of sensitive electronic components. So we may require additional filtering.

1.2.3 Dual Active Bridge Converter

A Dual Active Bridge is an isolated, bidirectional DC–DC converter in which both the input and output voltages are in DC form is shown in fig.1.3. It leverages two identical H-bridge configurations on both the primary and secondary sides, along with a high-frequency isolating transformer. The DAB converter incorporates active devices on both the input and output sides, resulting in a compact and efficient topology in which it helps

in minimizing the need for additional reactive components. It also ensures low stress on the devices and efficiently utilizes the transformer's leakage inductance as the primary energy transfer element. When the power is flowing from the DC side to the converter, this circuit acts as an inverter and then it feeds it to the transformer through additional inductor ie., leakage inductor and the transformer to the second part of the circuit, the second half of the circuit behaves like a rectifier, we are using a high-frequency transformer for high efficiency with electrical insulation. One major advantage of such a converter is the bidirectional power flow , efficient dc to dc energy transfer and ability to control the reactive power of the Grid.

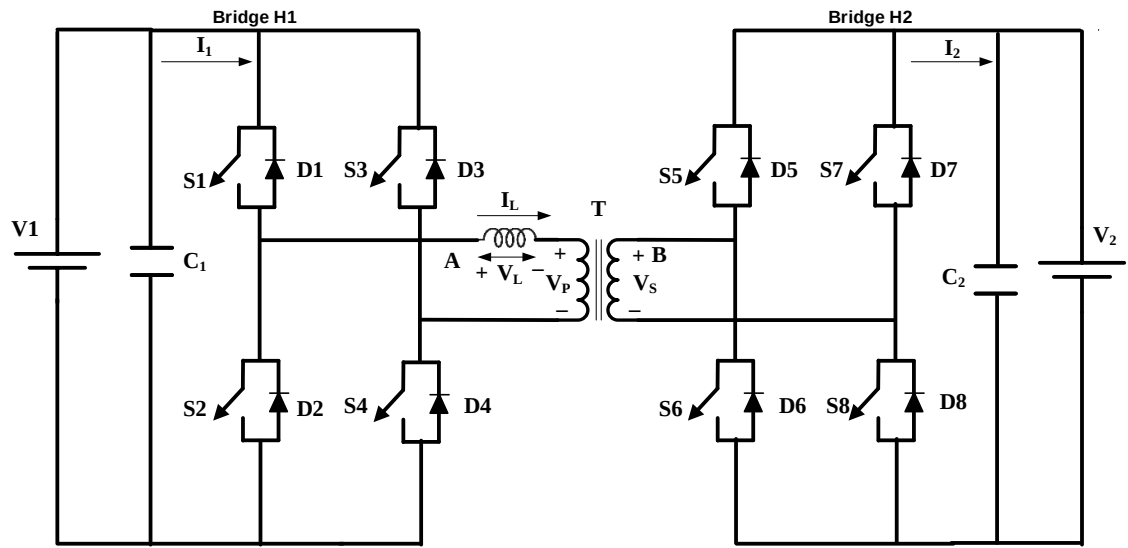


Figure 1.3: Circuit diagram of DAB Converter.

1.2.4 Power System Perspectives on the Dual Active Bridge

The two bridges of a DAB can be viewed as voltage buses linked through a reactive path. With the bridges represented as sources and the leakage inductance as the line reactance, the relative phase determines how much power is exchanged is seen in Fig.1.4

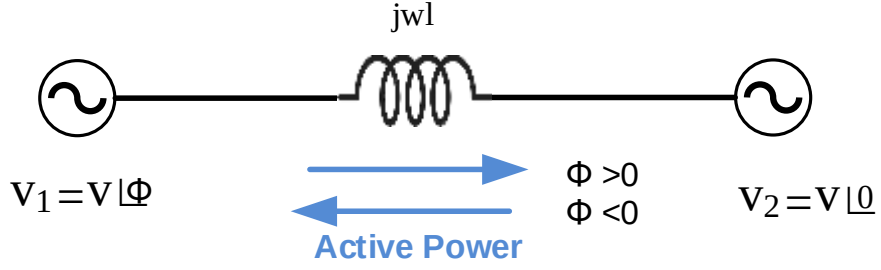


Figure 1.4: Power flow diagram between two sinusoidal voltage source.

1.3 Fundamental Analysis of the DAB

Let all phasors be RMS quantities. Denote the bridge output phasors as $V_a = V_1 \angle \delta_1$ and $V_b = V_2 \angle \delta_2$. The interconnection is modeled by a series inductive reactance jX_L with $X_L = \omega L$.

$$V_a = V_1 \angle \delta_1, \quad (1.2)$$

$$V_b = V_2 \angle \delta_2. \quad (1.3)$$

The current through the series inductance (taken from node a to node b) is

$$I_1 = \frac{V_a - V_b}{jX_L} = \frac{V_1 \angle \delta_1 - V_2 \angle \delta_2}{X_L \angle 90^\circ} \quad (1.4)$$

$$= \frac{V_1 \angle (\delta_1 - 90^\circ)}{X_L} - \frac{V_2 \angle (\delta_2 - 90^\circ)}{X_L}. \quad (1.5)$$

If a single fundamental frequency is present, the complex power relations are

$$S = V_0 * I_0^*, \quad (1.6)$$

$$P = VI \cos \theta, \quad (1.7)$$

$$Q = VI \sin \theta, \quad (1.8)$$

$$S = P + jQ. \quad (1.9)$$

Referencing complex power to bus b gives

$$S_b = V_b * I_1^* \quad (1.10)$$

$$= V_2 \angle \delta_2 * \left(\frac{V_1 \angle (\delta_1 - 90^\circ)}{X_L} - \frac{V_2 \angle (\delta_2 - 90^\circ)}{X_L} \right)^* \quad (1.11)$$

$$= \frac{V_1 * V_2}{X_L} \angle (90^\circ - \delta_1 + \delta_2) - \frac{V_2^2}{X_L} \angle 90^\circ. \quad (1.12)$$

Taking real and imaginary parts, with $\Delta\delta \triangleq \delta_1 - \delta_2$,

$$P_L = \frac{V_1 * V_2}{X_L} \cos(90^\circ - \Delta\delta) = \frac{V_1 * V_2}{X_L} \sin(\Delta\delta), \quad (1.13)$$

$$Q_L = \frac{V_1 * V_2}{X_L} \cos(\Delta\delta) - \frac{V_2^2}{X_L}. \quad (1.14)$$

Discussion. Equation (1.13) shows that net active power transfer depends on the phase difference $\Delta\delta = \delta_1 - \delta_2$; a nonzero phase shift causes power to flow from the leading bridge to the lagging bridge. This mechanism enables constant-frequency, square-wave operation while using the transformer leakage inductance (modeled as X_L) as the principal energy-transfer element. Resonant operation is avoided in this study to limit VA stress and simplify soft-switching control.

1.3.1 Characteristics of an Inductor Current

From the above inference obtained from equation (1.13), there must be a phase difference in the voltages across the inductor. Now analysing the inductor current behaviour. considering, $V_{in} > V_o$ and a unity turns ratio for the isolation transformer in Figx. Now, the voltage across the inductor is given by $V_L = V_{an} - V_{bn}$. During the period of t_1 when S_1, S_4 and S_6, S_7 are ON in the 1st and 2nd H bridge respectively, $V_L = V_{in} + V_o$ in magnitude and the hence the inductor current is having a higher gradient given by V_L/L . During the period t_2 when S_1, S_4 and S_5, S_8 are ON in both the 1st and 2nd H bridges respectively, $V_L = V_{in} - V_o$ in magnitude and the hence the inductor current is having a less gradient given by V_L/L compared to t_2 . During the period t_3 when S_2, S_3 and S_5, S_8 are ON in the 1st and 2nd H bridge respectively, $V_L = -V_{in} - V_o$ and the hence the inductor current is having a negative gradient in this interval. During the period t_4 when S_2, S_3 and S_6, S_7 are ON in both 1st and 2nd H bridge respectively, $V_L = -V_{in} + V_o$ and the hence the inductor

current is again having a negative gradient in this interval but it is less than t_3 . During the period t_5 when S_1, S_4 and S_6, S_7 are ON in both 1st and 2nd H bridge respectively, $V_L = V_{in} + V_o$ and hence the inductor current is now having a positive gradient in this interval. Thus we obtain the inductor current characteristic as shown in the figure.1.5

1.3.2 Operating Mode

Forward operation: In the forward operating mode the converter draws power from the voltage source. During each switching instant, two diagonally opposite switches conduct in each H-bridge. For converter 1 the conducting pair is S_1 and S_4 ; for converter 2 the conducting pair is S_5 and S_8 . At any instant four devices are in conduction is shown in fig.1.6.

Power flow sequence (forward):

1. Bridge 1: $V_1 \rightarrow S_1 \rightarrow \text{Inductor} \rightarrow \text{Transformer} \rightarrow S_4$.
2. Bridge 2: $\text{Transformer} \rightarrow S_5 \rightarrow \text{load} \rightarrow S_8 \rightarrow \text{Transformer}$.

Backward operation: In the reverse operating mode the power is taken from the load and returned toward the source. The conducting devices are S_7, S_6 in converter 2 and S_2, S_3 in converter 1. As before, four switches conduct at any given instant to maintain current flow is shown in fig.1.7.

Power flow sequence (reverse):

1. Bridge 2: $\text{load} \rightarrow S_7 \rightarrow \text{Transformer} \rightarrow S_4 \rightarrow \text{load}$.
2. Bridge 1: $\text{Transformer} \rightarrow S_3 \rightarrow V_1 \rightarrow S_2 \rightarrow \text{Transformer}$.

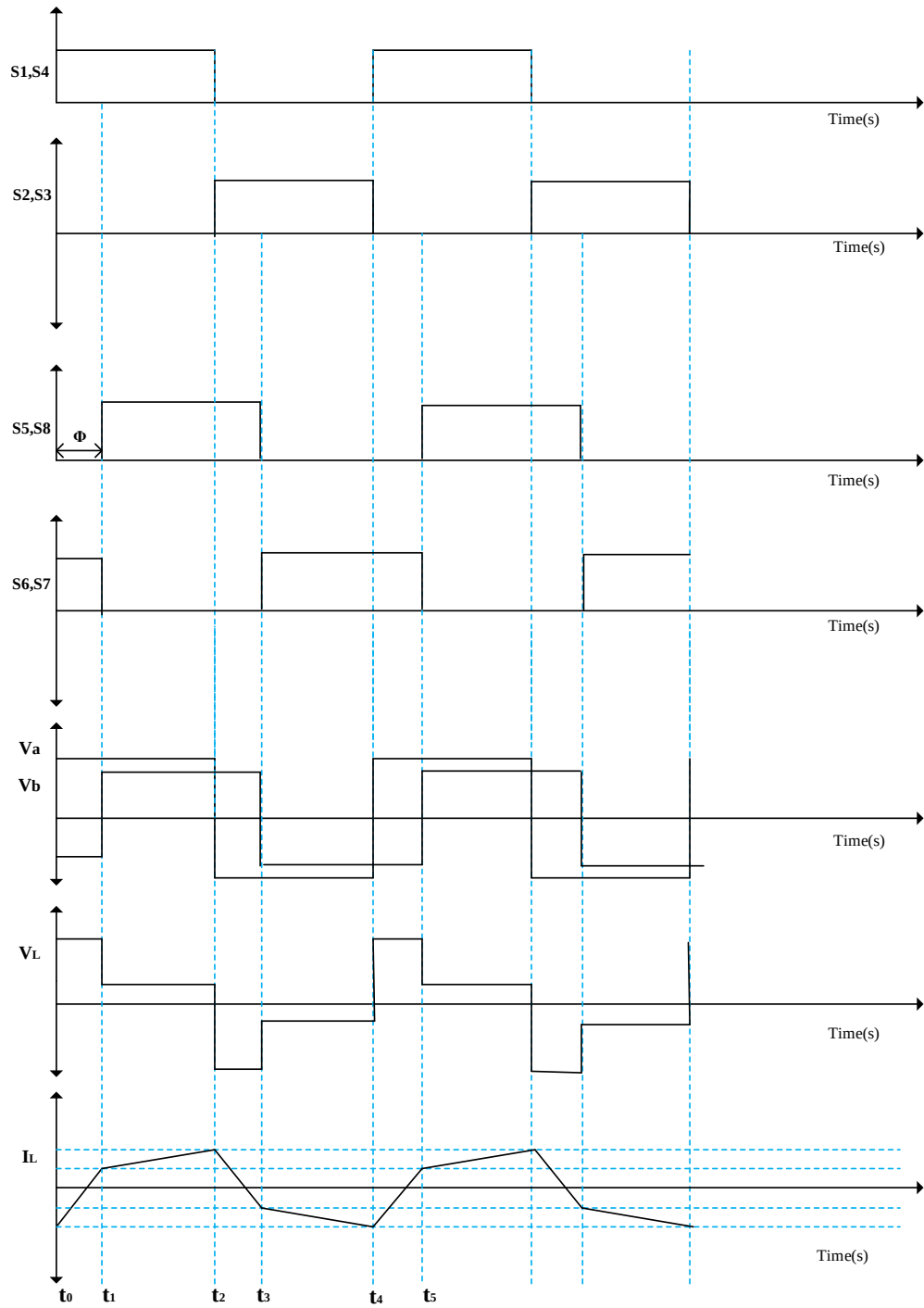


Figure 1.5: Wave forms of Dual Active Bridge Converter.

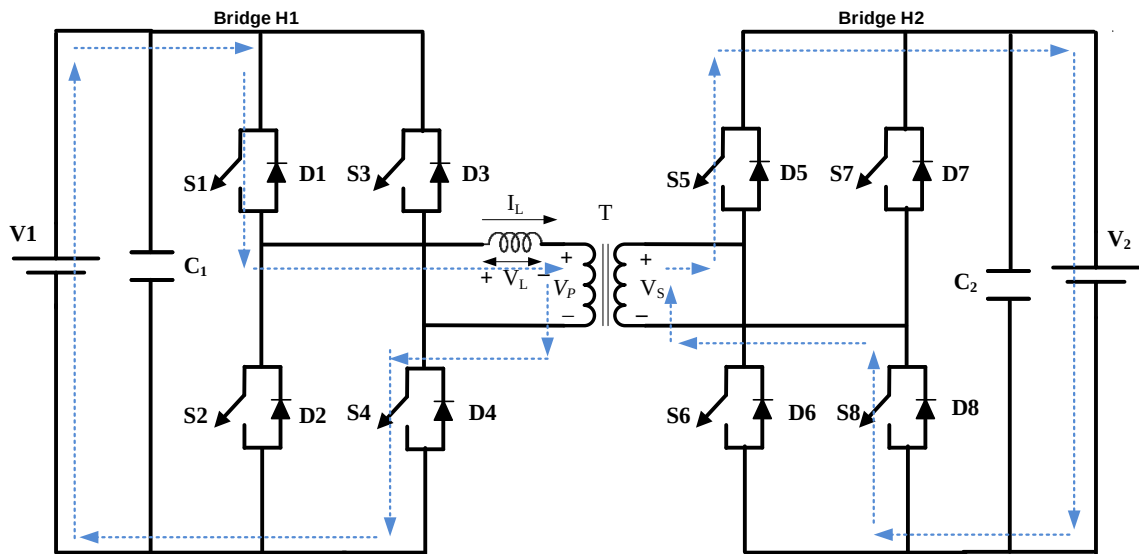


Figure 1.6: Converter Diagram of DAB in Forward Mode.

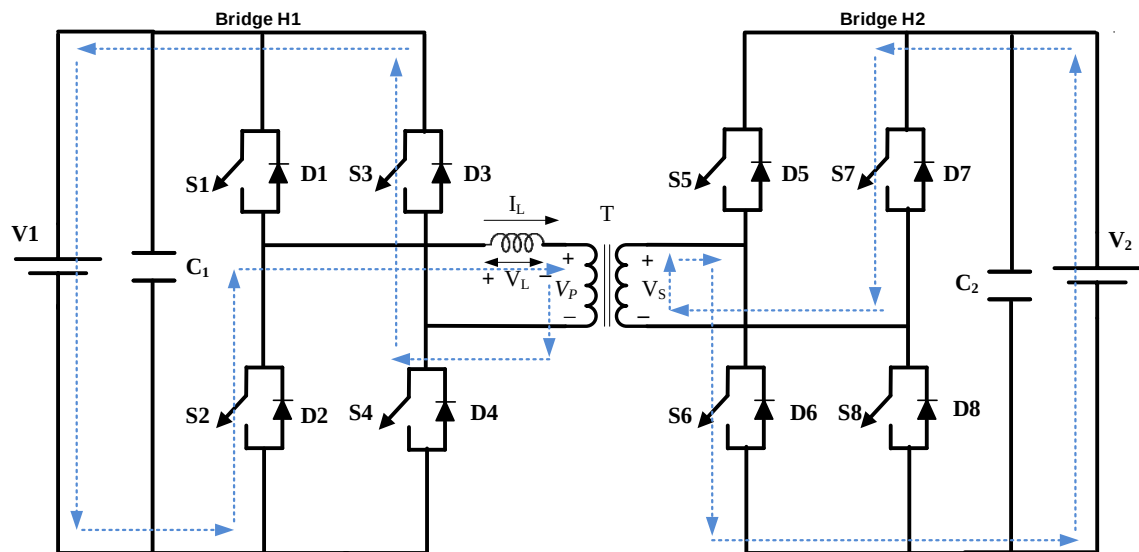


Figure 1.7: Converter Diagram of DAB in Backward Mode.

1.3.3 Mathematical expression of Dual Active Bridge Converter

during interval $t_0 - t_1$

$$i_L(t_1) = i_L(t_0) + (V_1 + V_2) * \frac{D * T_s}{L_s} \quad (1.15)$$

during interval $t_1 - t_2$

$$i_L(t_2) = i_L(t_1) + (V_1 - V_2) * \frac{(1 - D) * T_s}{L_s} \quad (1.16)$$

during interval $t_2 - t_3$

$$\Delta i_{L_s} = (V_1 + V_2) * \frac{-D * T_s}{L_s} \quad (1.17)$$

here we find the value of I_o

$$i_L(t_0) = \frac{(1 - 2D) * V_2 - V_1}{4 * f_s * L_s} \quad (1.18)$$

maximum value of inductor current

$$i_L(t_1) = I_{L,\max} = \frac{-(1 - 2D) * V_1 + V_2}{4 * f_s * L_s} \quad (1.19)$$

average value of inductor current

$$I_{L,\text{avg}} = \frac{D * (1 - D) * V_2}{2 * f_s * L_s} \quad (1.20)$$

input power

$$P_{\text{in}} = n * V_1 * I_{L,\text{avg}} = n * V_1 * \frac{D * (1 - D) * V_2}{2 * f_s * L_s} \quad (1.21)$$

since the DAB is DC-DC, output power is also DC

$$P_{\text{in}} = P_{\text{out}} = \frac{V_2^2}{R} \quad (1.22)$$

from the power balance we can find the output voltage V_2

$$V_2 = n * V_1 * R_L * \frac{D * (1 - D) * V_2}{2 * f_s * L_s} \quad (1.23)$$

Chapter 2

Phase Shift Control and Reactive Power in DAB Converter

2.0.1 Control Techniques

[6]Majorly four types of control principal were used in Dual Active Bridge Converter, in control techniques we mainly control the phase shift between the output voltage of two bridge converts so that the power flow can be possible from the leakage inductance of transformer also it is know that in AC system power flow one end to other end when it is change in voltage angle.

2.0.2 Single Phase Shift (SPS)

In single phase shift control technique, in converter1 the opposite leg switches in going in conduction mode and we give the phase shift in 2nd bridge converter for the power flow in from one end to other end, as it is easiest and most useable technique in DAB the disadvantage of this technique as is power flow depend on leakage inductance so circulating current so we dont have full range of ZVS and get the 2 level output voltage.

2.0.3 External Phase Shift (EPS)

In this control scheme, cross connected switch of one converter is in conduction mode, while in another converter switches are inner phase shift ratio, so output voltage of one converter is three level where as other one (0.5) duty cycle square wave as we know that

outer phase shift for power flow where as inner phase shift for obtain the full range of ZVS and reduce the circulating current.

2.0.4 Dual Phase Shift (DPS)

[7]As compared to EPS control scheme in this control scheme cross connected switches in each full bridge converter has inner phase shift so it provide more wide range for ZVS and inner phase shift for each full converter is same as the both converter are same so power flow direction can be changed so it is more perfer over EPS. the output voltage level for both full converter is 3 level.

2.0.5 Triple Phase Shift (TPS)

The only difference between DPS and TPS the inner phase shift between two bridge converter is not equal.

2.0.6 Comparitive analysis of SPS and DPS of DAB Converter and it's reactive Power elimination

2.0.7 Single-Phase-Shift Operating Characteristics

The single phase shift strategy has been studied extensively in the literature [1]. Figure 1.3 sketches the fundamental bridge voltages and the inductor current segments that arise under SPS. At each commutation instant, the inductor current assumes well-defined values that follow directly from the applied bridge voltages and the series (leakage) inductance. The current equations were

$$I_1 = \frac{T_{hf}}{2L} [nV_o(2d - 1) + V_i] \quad (2.1)$$

$$I_2 = \frac{T_{hf}}{2L} [nV_o - V_i(2d - 1)] \quad (2.2)$$

where $T_{hf} = T/2$ and d denotes the outer phase-shift ratio between bridges (as depicted in Fig. 2). The above expressions correspond to forward power transfer and yield the familiar quadratic dependence of power on phase shift. In steady state, the average output power is

$$P_o = \frac{nV_iV_oT_{hf}}{L} (1 - d) d. \quad (2.3)$$

Equations (2.1)–(2.3) emphasize two practical points: (i) phase shift directly sets the inductor current swing and thus the direction and magnitude of power flow, and (ii) for fixed voltages, the achievable power scales inversely with the leakage inductance L , which also sets the current ripple.

2.0.8 Alternative Closed-Form Relations for SPS

The following steady-state relations (expressed for a single switching period) are also widely used to parameterize SPS behavior. As reported in [4], the maximum transformer current $I_{\max}$ and the initial secondary current I_0 are

$$I_{\max} = \frac{n}{4f_s L_s} \left[- (1 - 2D) nV_1 + V_2 \right], \quad (2.4)$$

$$I_0 = \frac{n}{4f_s L_s} \left[(1 - 2D) V_2 - nV_1 \right], \quad (2.5)$$

where f_s is the switching frequency, L_s the leakage inductance, D the outer phase-shift ratio (D is often used interchangeably with d in the SPS literature), V_1 and V_2 the bridge voltages, and n the turns ratio.

These expressions make clear that any mismatch between the reflected voltages ($V_2 \neq nV_1$) creates a baseline transformer current even at zero phase shift ($D = 0$, i.e., zero net power), and conversely that exact volt-second balance ($V_2 = nV_1$) is the only condition that eliminates this offset at $D = 0$. The resulting offset current is non-working: it circulates energy within a period and appears as reactive current.

2.0.9 Reactive Power Under SPS

Unlike unidirectional PWM converters (buck/boost/buck-boost), where power delivered each cycle is predominantly real, the DAB applies square voltages on both sides of the transformer and couples them through the leakage inductance. The phase of current therefore need not coincide with the voltage phase, and a portion of the energy exchanged

within each switching period is returned to the source rather than delivered to the load. Defining the reactive component over one period as

$$Q = \frac{1}{T_s} nV_1 \int i(t) dt \quad (2.6)$$

with $T_s = 1/f_s$ and $i(t)$ the transformer current, a closed-form expression for Q is

$$Q = nV_1 \frac{[(1 - 2D)V_2 - nV_1]^2}{16 f_s L_S} \times \begin{cases} \frac{1}{nV_1 + V_2}, & I_0 < 0, \\ 0, & I_0 = 0, \\ \frac{1}{nV_1 - V_2}, & I_0 > 0. \end{cases} \quad (2.7)$$

The sign of Q conveys the current–voltage phasing: $Q > 0$ corresponds to lagging current (inductive behavior) typical of heavier load points where $I_0 < 0$, whereas $Q < 0$ indicates leading current (capacitive behavior) common at light load where $I_0 > 0$. Except along the locus $(1 - 2D)V_2 = nV_1$, nonzero reactive power is present and grows more pronounced as load decreases. Practically, this circulating component elevates RMS device current and conduction loss, which motivates phase-shift refinements (e.g., dual/extended phase shift) when high efficiency is required at reduced output power.

2.0.10 Dual Phase Shift of Dual Active Bridge Converter:

[1] In dual-phase-shift control, an *inner* phase displacement D is introduced symmetrically within both bridges, in addition to the *outer* phase displacement d between the bridges is shown in fig.2.1. To avoid overlapping segments within a half switching period, they must satisfy the condition

$$0 \leq d \leq 1 \quad 0 \leq D \leq 1 \quad 0 \leq d + D \leq 1$$

The following conditions will guarantee the pulse blocks fit within the high-frequency window without violating causality or dead-time constraints.

Operating modes: Two operating regimes arise depending on the relative magnitudes of the outer and inner shifts. When $|d| \geq D$, the inner shift sits entirely within the outer window; when $|d| < D$, the inner shift dominates and effectively clips the outer window. The corresponding closed-form expressions for the average output power are:

$$P_o = \frac{nV_iV_oT_{hf}}{L} \left[(1-d)d - \frac{1}{2}D^2 \right] \quad \text{for } |d| \geq D, \quad (2.8)$$

$$P_o = \frac{nV_iV_oT_{hf}}{L} d \left[1 - D - \frac{1}{2}d \right] \quad \text{for } |d| < D. \quad (2.9)$$

Current shaping and trade-offs: Operating with $|d| \leq D$ prioritizes current shaping, the RMS inductor current is reduced because the inner shift trims circulating intervals that do not contribute to net transfer. The penalty is a lower delivered power for the same d compared with the $|d| \geq D$ case. At a fixed output power, however, DPS achieves a smaller peak current than SPS, which directly lowers conduction stress and magnetic ripple.

Reactive power behavior. Relative to single-phase-shift, the additional degree of freedom D lets the controller suppress the non-working (reactive) component of the inductor current. In practical terms, DPS curtails the energy shuttled back and forth within a switching period while preserving the commanded active transfer. The outcome is a lower RMS current for the same P_o , improved soft-switching margins at reduced load, and a more benign thermal profile for the semiconductor devices.

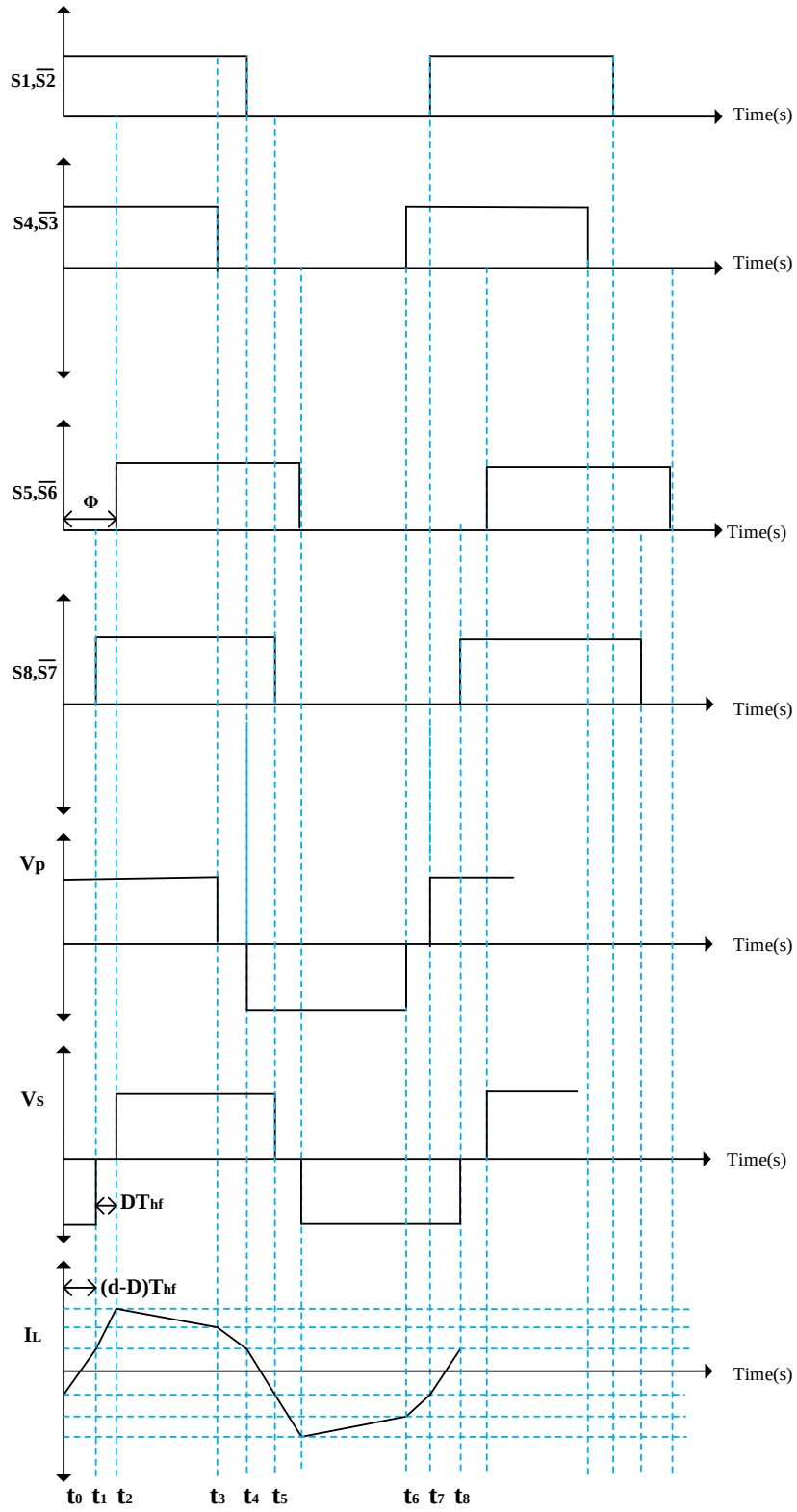


Figure 2.1: Wave forms of DPS DAB Converter.

Chapter 3

Soft Switching Analysis

3.0.1 Zero Voltage Switching Analysis on Dual Active Bridge Converter

[8] Every leg in the bridge uses a brief dead time between turning one MOSFET off and the other on. During this period of time, the energy stored in the leakage/series inductance keeps the current flowing and shuffles charge between the MOSFET output capacitances. If that current has the right direction and enough magnitude, it pulls the voltage on the incoming device down to (near) zero before its gate is driven, then the voltage across the MOSFET is close to zero at turn on, is called zero voltage switching (ZVS).

While the primary keeps conducting through S1 and S4, the secondary is about to swap states: S6 and S7 are turned off, and S5 and S8 are the next pair to turn on. At the start of dead time, all four secondary devices are off. The inductor current then naturally discharges the capacitances of S5 and S8 and charges those of S6 and S7 up to the secondary bus. The current commutation is shown in Figure.3.1

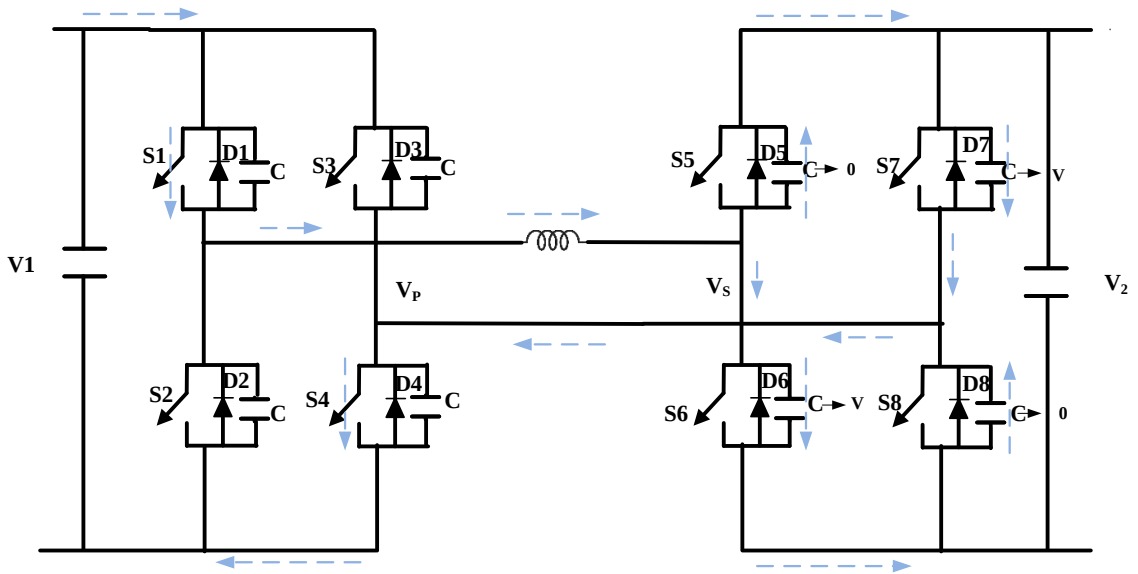


Figure 3.1: Converter Diagram ZVS Transition on Secondary Side-Capacitor.

As soon as the node voltages at S5 and S8 fall close to zero, their body diodes D5 and D8 take over the current path, clamping the device voltages at 0 V. With the nodes clamped, S5 and S8 can now be gated on at zero voltage, cutting turn-on loss. (The arrows near the diodes indicate that the diode is conducting while the MOSFET channel is still off.)

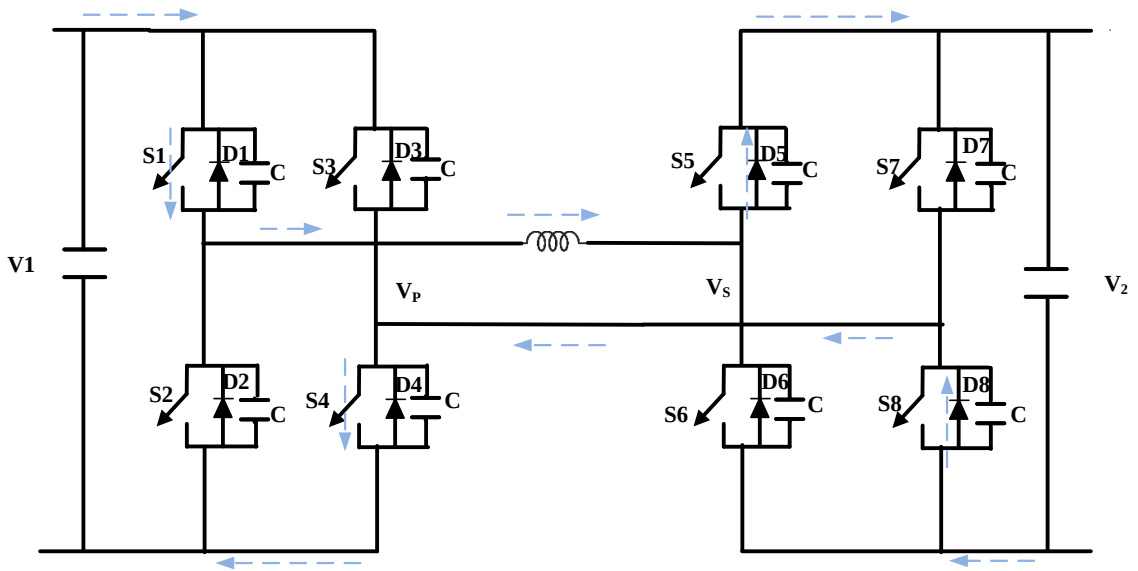


Figure 3.2: Converter diagram Converter Diagram ZVS Transition on Secondary Side-Diode.

The same same scenario happens out on the primary in the next transition. The secondary keeps conducting through S5 and S8, while the primary leg changes over: S1 and S4

turn off; S2 and S3 are scheduled to turn on. During the primary dead time, the inductor current discharges the capacitances of S2 and S3 and charges those of S1 and S4 toward the primary bus.

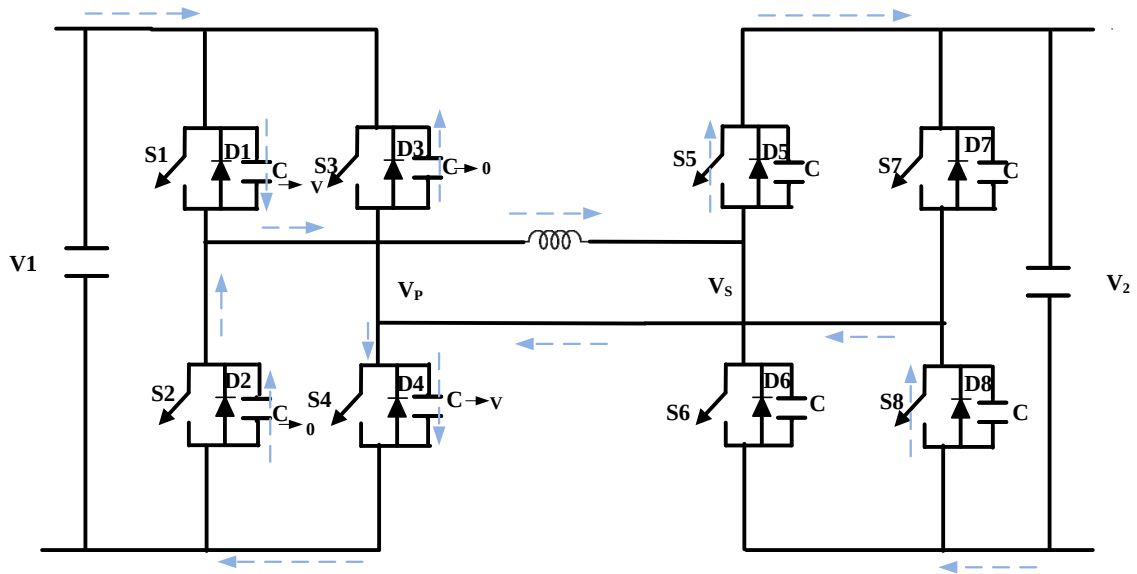


Figure 3.3: Converter Diagram ZVS Transition on Secondary Side-Capacitor.

Once the voltages at S2 and S3 collapse, body diodes D2 and D3 conduct, clamping their node voltages and paving the way for true ZVS when S2 and S3 receive their gate signals.

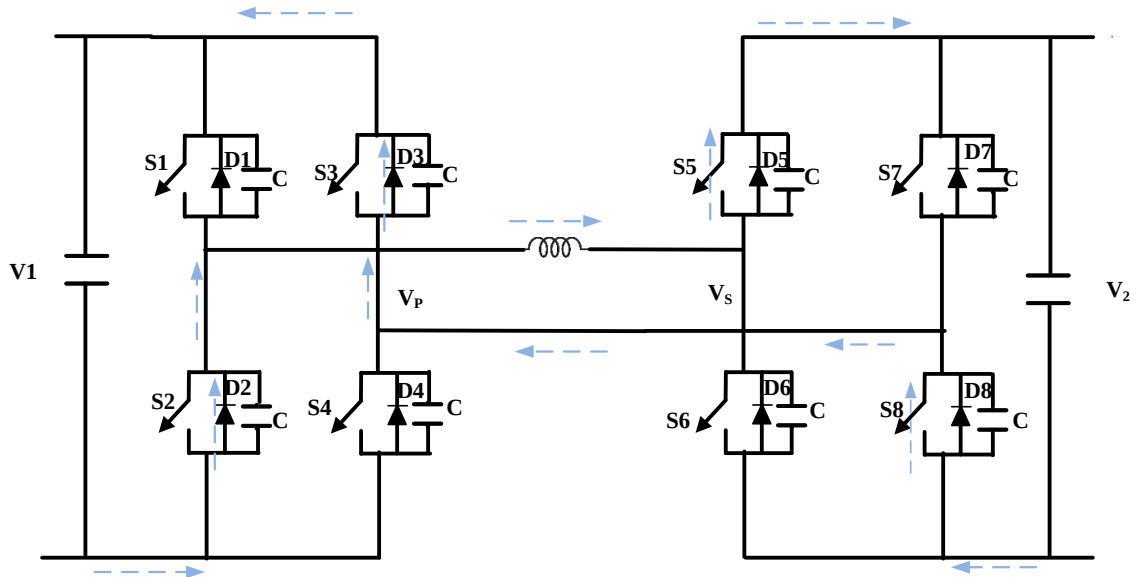


Figure 3.4: Converter Diagram ZVS Transition on Secondary Side-Diode.

Chapter 4

Simulation and Results

4.1 Simulation and Calculation results

Single Phase Shift (SPS) of DAB — Calculations

Parameter values

$$V_1 = 800 \text{ V}, \quad V_2 = 800 \text{ V}, \quad f_s = 100 \text{ kHz}, \quad n = 1, \quad P_o = 30 \text{ kW}.$$

Calculations

$$R_{\text{load}} = \frac{V_2^2}{P_{\text{out}}} = \frac{800^2}{30 \times 10^3} = 21.33 \, \Omega, \quad (1)$$

$$(I_{\text{in}})_{\text{avg}} = \frac{P_{\text{out}}}{V_1} = \frac{30 \times 10^3}{800} = 37.5 \text{ A}, \quad (2)$$

$$(I_{\text{out}})_{\text{dc}} = \frac{P_{\text{out}}}{V_2} = \frac{30 \times 10^3}{800} = 37.5 \text{ A}. \quad (3)$$

$$\text{Voltage ripple } \Delta V = 3\% \cdot V_2 = 24 \text{ V},$$

$$C = \frac{I_o T_s}{\Delta V} = \frac{37.5 \times 10 \, \mu\text{s}}{24} = 1.5625 \times 10^{-5} \text{ F} \approx 15.6 \, \mu\text{F}. \quad (4)$$

$$L_{\text{req}} = \frac{V_1 V_2}{8 n f_s P_{\text{out}}} = \frac{800 \cdot 800}{8 \cdot 100 \times 10^3 \cdot 30 \times 10^3} = 2.666 \times 10^{-5} \text{ H} \approx 26.66 \mu\text{H}, \quad (5)$$

$$L_{\text{act}} = 0.8 L_{\text{req}} \approx 21.33 \mu\text{H}. \quad (6)$$

For SPS with phase angle δ (rad) between bridge fundamentals,

$$P_o = \frac{V_1 V_2}{2 n L_{\text{act}} f_s \pi^2} \delta(\pi - \delta). \quad (8)$$

$$k \equiv \frac{2 n L_{\text{act}} f_s \pi^2}{V_1 V_2} P_o, \quad \Rightarrow \quad \delta(\pi - \delta) = k.$$

With $L_{\text{act}} = 21.33 \mu\text{H}$ and $P_o = 30 \text{ kW}$,

$$k = \frac{2 \cdot 1 \cdot 21.33 \times 10^{-6} \cdot 10^5 \cdot \pi^2}{800 \cdot 800} \cdot 3 \times 10^4 = 1.9736. \quad (9)$$

Hence,

$$\delta^2 - \pi\delta + k = 0 \Rightarrow \boxed{\delta = \frac{\pi - \sqrt{\pi^2 - 4k}}{2}} = \frac{\pi - \sqrt{\pi^2 - 4 \cdot 1.9736}}{2} = 0.868 \text{ rad} \approx 49.75^\circ. \quad (10)$$

4.2 Open Loop and Closed Loop Simulation of a Dual Active Bridge Converter

4.2.1 Model used in both cases

The converter is modeled with two full H-bridges, an ideal high-frequency transformer with turns ratio n , and a series/leakage inductance L_s that carries the transferred power. Comparison between single Phase Shift (SPS) and Dual Phase shift (DPS) of DAB is also done for studying circulating current.

4.2.2 Open Loop Simulation of (SPS) DAB Converter

Validate power transfer physics and observe inductor-current shaping without any feedback action. The outer phase shift $d \in [0, 0.5]$ (and the inner shift D in DPS) is imposed as a fixed parameter.

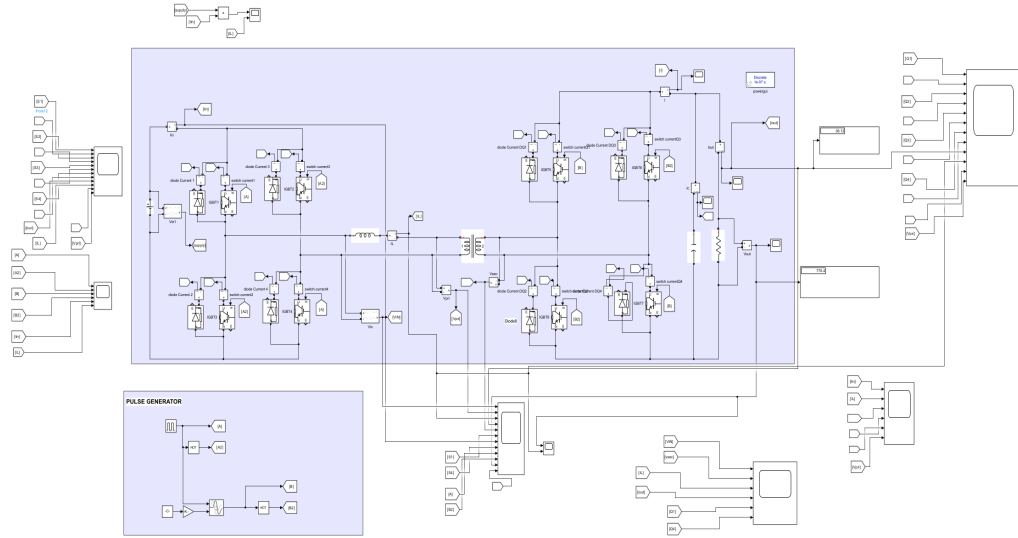


Figure 4.1: Matlab Simulink - DAB Open loop.

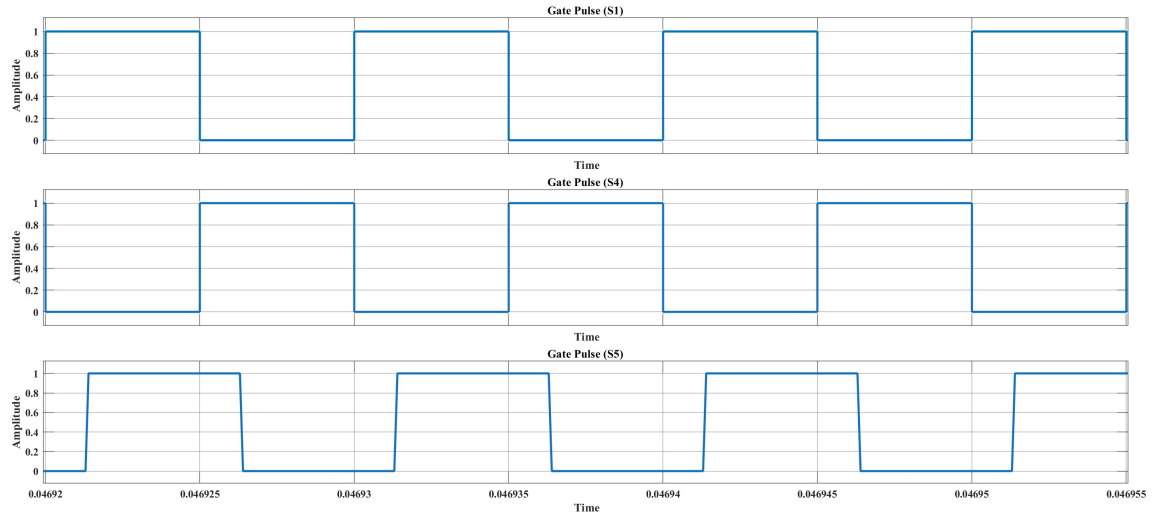


Figure 4.2: Matlab Simulink Waveforms - Gate Pulse S1,S4,S5.

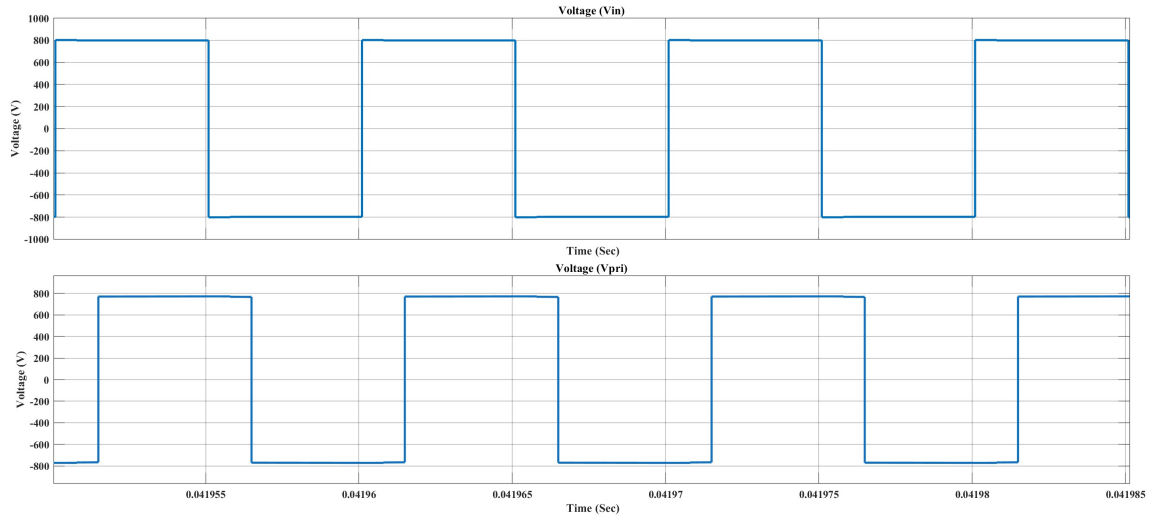


Figure 4.3: Matlab Simulink Waveforms - V_{in}, V_{pri} .

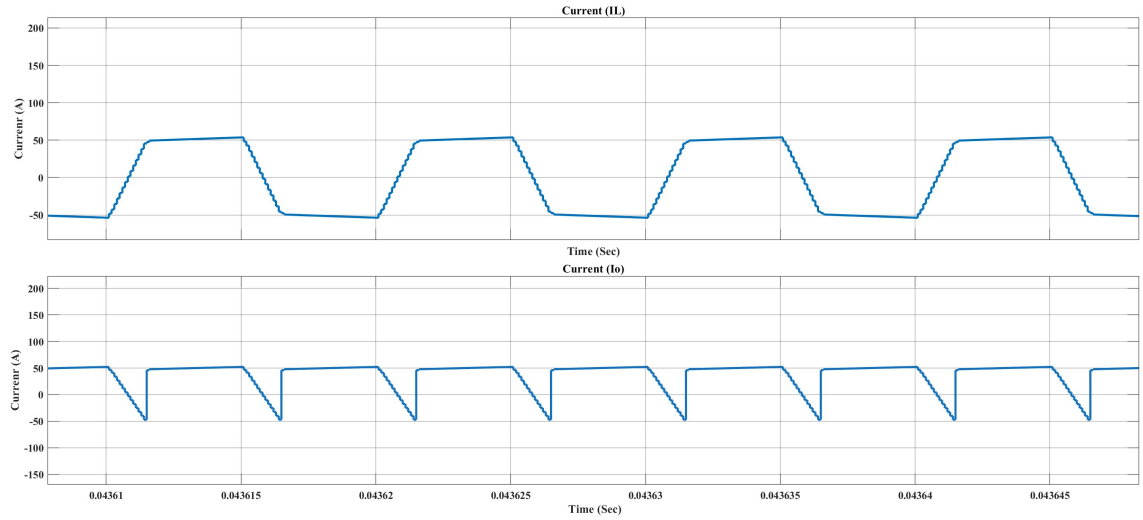


Figure 4.4: Matlab Simulink Waveforms - I_L, I_o .

4.2.3 Closed Loop simulation of (SPS) DAB Converter

Regulate a target variable (typically the secondary DC voltage or transferred power) while minimizing RMS current and preserving ZVS across operating conditions. Here it regulates the secondary DC voltage V_2 (or power) with a PI controller with hit and trail method such that it provided with required phase shift value provided to the bridge-2 switches. The required values of PI controller were provided below.

Controller	Role	K_p	K_i
$PI_1(s)$	Outer voltage loop	0.06	0.35
$PI_2(s)$	Inner loop	0.0009	0.75

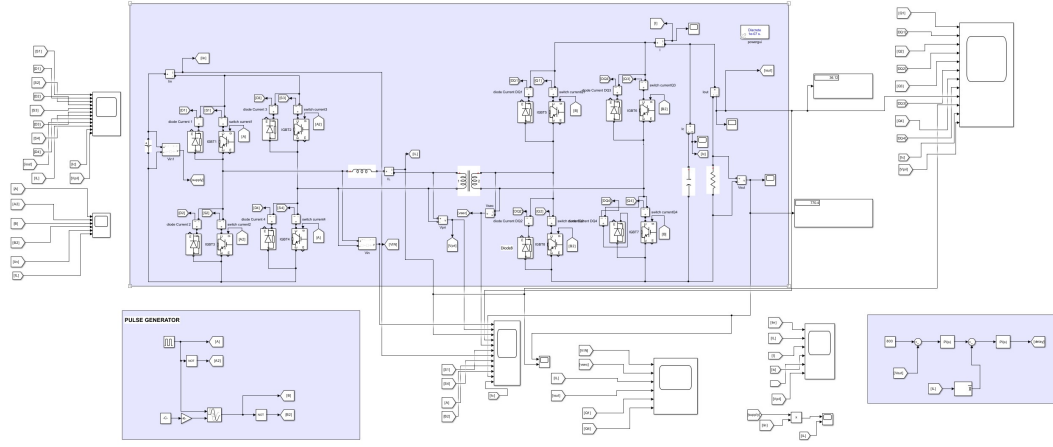


Figure 4.5: Matlab Simulink - DAB Closed loop.

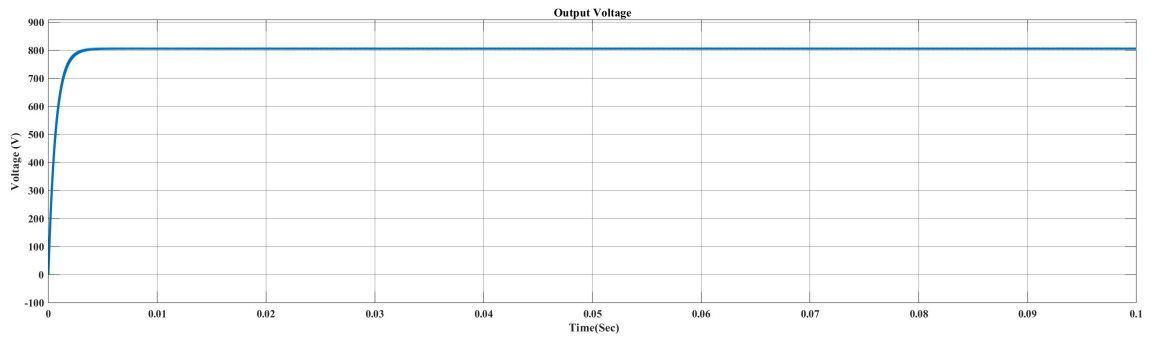


Figure 4.6: Matlab Simulink - DAB Closed loop.

Dual Phase Shift (DPS) of DAB — Calculations

Same parameter values

$$V_1 = 800 \text{ V}, \quad V_2 = 800 \text{ V}, \quad f_s = 100 \text{ kHz}, \quad L = 21.33 \mu\text{H}, \quad n = 1, \quad P_o = 30 \text{ kW}.$$

Outer phase-shift ratio: d . Inner phase-shift ratio: D . Angle forms: $d = \frac{\delta}{\pi}$, $D = \frac{\beta}{\pi}$.

Case (i): $|d| \geq D$

$$P_o = \frac{nV_1V_2}{2Lf_s} \left[d(1-d) - \frac{1}{2}D^2 \right] = \frac{nV_1V_2}{2Lf_s} \left[\frac{\delta}{\pi} \left(1 - \frac{\delta}{\pi} \right) - \frac{1}{2} \left(\frac{\beta}{\pi} \right)^2 \right]. \quad (4.1)$$

Compute the normalization constant

$$k \equiv \frac{2Lf_sP_o}{nV_1V_2} = \frac{2(21.33 \times 10^{-6})(1 \times 10^5)(3 \times 10^4)}{800 \cdot 800} \approx 0.2000.$$

Using the SPS nominal phase (from your earlier sheet) $\delta = 0.868 \text{ rad } (49.75^\circ)$,

$$\frac{\delta}{\pi} \left(1 - \frac{\delta}{\pi} \right) \approx 0.1998 \approx k.$$

Thus the bracket in (4.1) must stay equal to k , which gives a very small inner shift:

$$\boxed{\beta \approx 0.01010 \text{ rad } (0.579^\circ)}$$

Case (ii): $|d| < D$

$$P_o = \frac{nV_1V_2}{2Lf_s} d \left(1 - D - \frac{1}{2}d \right) = \frac{nV_1V_2}{2Lf_s} \frac{\delta}{\pi} \left(1 - \frac{\beta}{\pi} - \frac{\delta}{2\pi} \right). \quad (4.2)$$

Imposing the same P_o (so the left-hand side again equals k), and choosing $D > d$ (inner shift larger than outer shift), one solution from the notes is

$$\boxed{\beta \approx 0.4905 \text{ rad } (28.1^\circ)}.$$

4.2.4 Open Loop Simulation of (DPS) DAB Converter

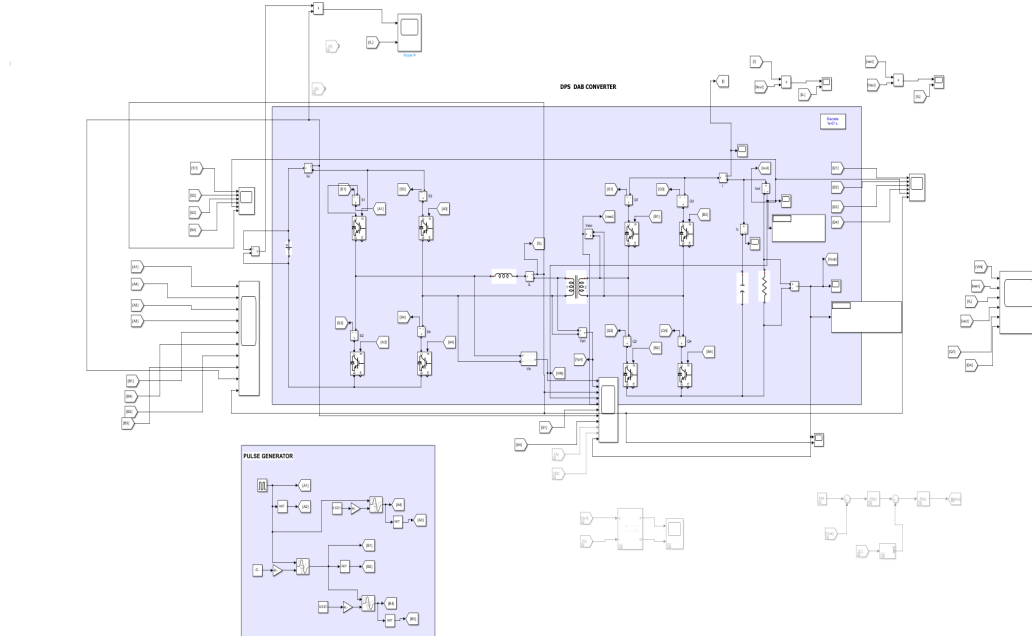


Figure 4.7: Matlab Simulink - DPS DAB Open loop.

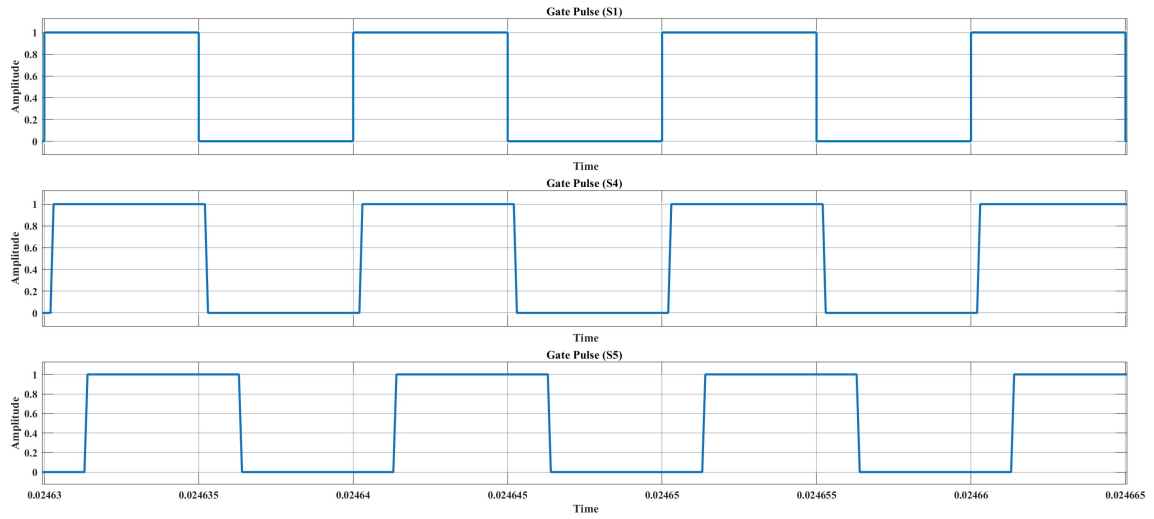


Figure 4.8: Matlab Simulink - DAB Gate Pulse S1,S4,S5.

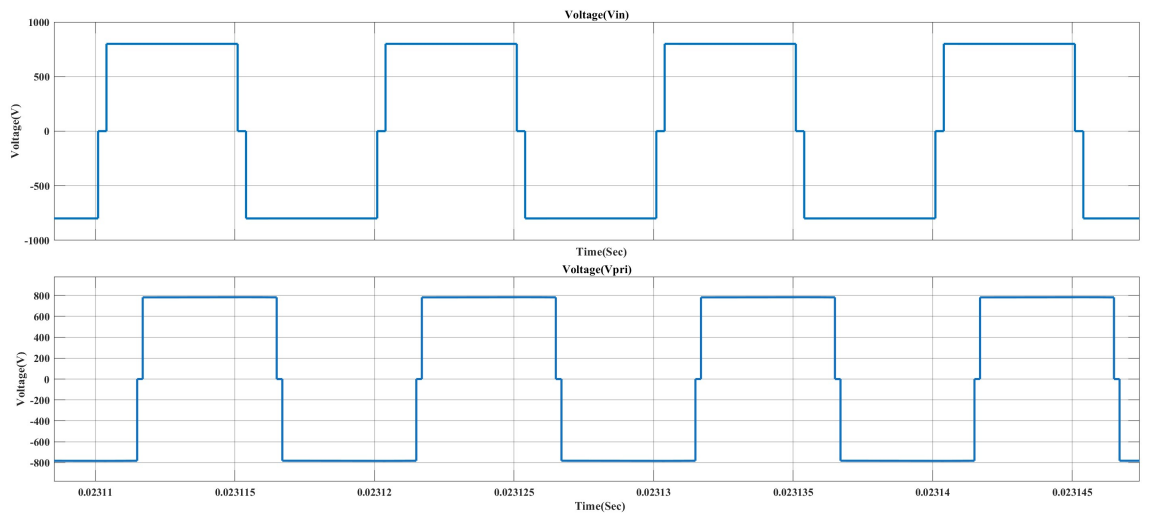


Figure 4.9: Matlab Simulink Wveforms - DPS DAB Voltage Vin,Vpri.

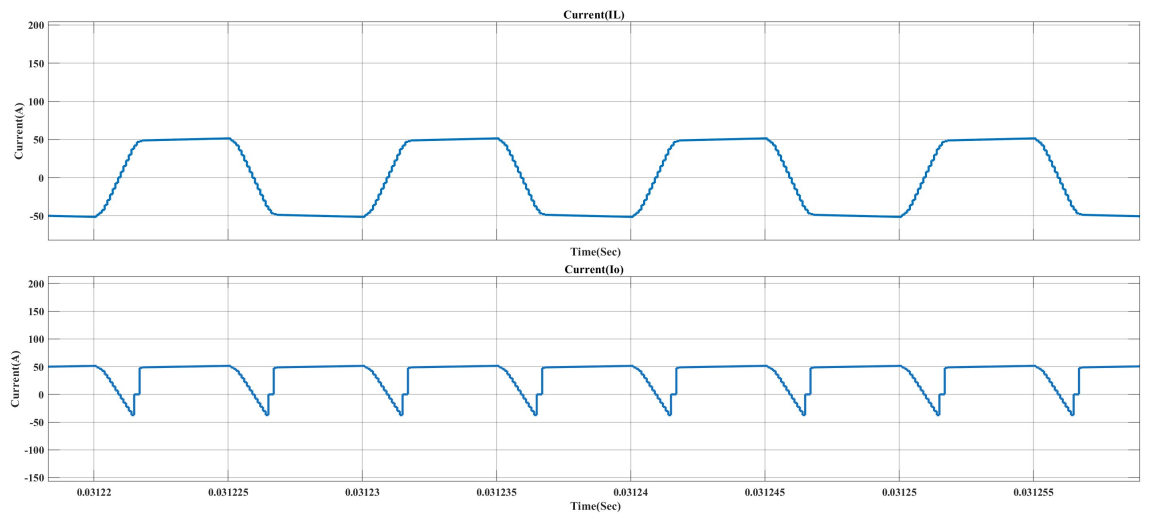


Figure 4.10: Matlab Simulink Waveforms - DPS DAB Current IL,Ipri.

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

The DAB's fixed-frequency, phase-shift power modulation makes it a compact and robust choice for isolated, bidirectional conversion. SPS establishes the fundamental behavior and is simple to commission in PSIM, but it carries reactive current at light load and narrows ZVS. Introducing DPS (and related inner-shift strategies) demonstrably reduces that reactive component and extends ZVS, improving efficiency and device utilization without changing magnetics or base frequency. For the nominal 800 V/800 V, 100 kHz, 30 kW operating point studied here, a leakage/series inductance on the order of tens of microhenries enables practical phase angles while keeping current ripple and device stress within bounds. These choices translate directly into PSIM scope the leakage current (triangular/trapezoidal), confirm soft turn on at device nodes, and check port power balance over a steady condition, when these signatures match theory, the model is ready for controller in the loop experiments

5.2 Future Scope

The Future work will focus on power stage analysis of a three winding, quad port dual active bridge converter. We will apply the same methodology used for the two winding DAB, then implement the three winding quad port dual active bridge in hardware to validate performance under real operating conditions. A closed loop controller will be developed to improve voltage regulation and dynamic response. Finally, adding bidirectional char-

ging capability will make the system suitable for future smart grid environments. Reactive power aware control at scale: Implement unified or optimized DPS variants that schedule inner outer shifts online to minimize RMS current across operating points (including light load standby), then validate efficiency gains in PSIM before hardware.

From 2-port to 3-port: Extend the model to a triple-active-bridge (TAB) for simultaneous routing between a stiff DC bus, a battery, and a fast buffer (e.g., supercap). Investigate phase-coordinated power sharing and ZVS boundaries across all three ports.

Application alignment: Map operating regions to specific use-cases (EV DC link, drilling-rig auxiliary DC bus, more-electric aircraft) and quantify benefits versus alternative isolated topologies using the same PSIM harness.

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